THE STATE-SPACE COMPLEXITY OF MALAWIAN BAWO: A COMBINATORIAL UPPER BOUND

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ABSTRACT

Bawo is the Malawian variant of the four-row mancala game Bao, played competitively by the Yao. While the Zanzibar (Swahili) form of Bao has received sustained attention in computational game theory, the Malawian variant has not, to our knowledge, been formally specified or measured. This paper takes a first step. We give a finite-state model of Bawo, separate the configuration space from the legal and reachable state spaces, and derive the exact cardinality of the configuration space, which upper-bounds both. The bound is approximately 1.7×10^{26} configurations, and exactly half this, about 8.7×10^{25}, once a conditionally stated player-swap symmetry is quotiented out. This figure is an upper bound only: it places Bawo no higher than the chess range and well below Go, but because most configurations are unreachable under legal play it leaves the lower end of Bawo's true position count open, pending a reachability estimate that we specify but do not yet carry out. We are careful to distinguish what is unconditional (the configuration count) from what depends on a not-yet-formalised rule set (position adequacy, transition totality, and symmetry invariance), and we set out the empirical methodology by which the reachable state space, the quantity of ultimate interest, is to be estimated. All figures are reproducible from the short script in Appendix A.

Keywords Bawo · Bao · mancala · state-space complexity · combinatorial enumeration · game complexity

1 Introduction

Bao is a member of the mancala family of count-and-capture board games and is widely regarded as the most complex of them in terms of rule structure and strategic depth. It is played on a 4×8 board with 64 seeds. In Malawi a close variant is played by the Yao under the name Bawo, contested at district, regional, and national levels.

Classical board games are routinely characterised by their *state-space complexity*, the number of legal positions reachable from the initial setup, and by their *game-tree complexity*, the number of distinct playable games. Such measures have been computed or estimated for chess, checkers, Go, and several mancalas, and they inform the design of search algorithms and solvers. For Bao, this work has concentrated on the Zanzibar form. De Voogt formalised its rules and studied human mastership [3]; Donkers, Uiterwijk, and de Voogt analysed its game properties and solved its novice version [4]; and the termination of individual moves was studied under the heading of never-ending moves [6].

The Malawian variant has not received comparable treatment. It differs from the Zanzibar form in several rules, most relevantly here the opening configuration of the *kuu* (the special pit, called *nyumba* in Swahili), the size of the opening reserve, and an explicit procedural rule for halting a move that risks not terminating. The purpose of this paper is to begin a state-complexity analysis specific to this variant, while being explicit about which claims are unconditional and which await a formal rule specification. We regard the unconditional configuration bound, together with the model that isolates the assumptions a rule specification must later discharge and the methodology that will consume it, as a self-contained increment worth recording before the heavier empirical work it scaffolds.

Contributions.

1. A finite-state model of Bawo that separates the configuration space C, the legal state space S, and the reachable state space S_{reach}, with $S_{\text{reach}} \subseteq S \subseteq C$ (Section 2).
2. The exact cardinality of C, derived by phase with an explicit partition variable and an injectivity argument that rules out double counting (Section 3, Theorem 2). As $S_{\text{reach}} \subseteq C$, this is an upper bound on the state-space complexity.
3. An explicit accounting of the assumptions on which the model rests (position adequacy, transition totality, symmetry invariance), each isolated so that the configuration count remains valid independently of them (Assumptions 1 and 2, Remark 2).
4. A placement of Bawo among other games (Section 4) and a statement of the methodology for estimating $|S_{\text{reach}}|$ (Section 5).

Scope. We analyse the competitive Malawian rules as reported in the secondary literature. Where these are underdetermined or regionally variable (notably the exact capture and relay conditions, and whether the opening reserve is 18 or 20 seeds), we parameterise rather than fix the rule, and we mark every such dependency. A fully executable rule specification, required for the reachability estimate of Section 5 and for discharging the assumptions of Section 2, is left to companion work.

2 Configuration, legal, and reachable states

2.1 Board, seeds, and phases

The board consists of 32 pits arranged in four rows of eight. The two near rows belong to one player and the two far rows to the other. Let $P = \{0, 1, \ldots, 31\}$ index the pits. There are 64 indistinguishable seeds, 32 originating with each player. Play proceeds in two phases. In the opening *namua* phase each player introduces seeds, one per turn, from a private reserve. The Malawian opening places either eight or ten seeds in each player's kuu, leaving a reserve of 20 or 18 respectively. Since each player owns 32 seeds, the remaining four per player occupy designated pits in the opening layout; the configuration count of Section 3 depends only on the reserve cap and not on where those seeds sit. When both reserves are exhausted the game enters the *mtaji* phase, in which all 64 seeds are on the board and moves redistribute seeds already in play. (The word *mtaji* also names a capturing move; we use it here only for the phase.)

2.2 Positions and the configuration space

Definition 1 (Position)**.** A position is a tuple $s = (B, r_A, r_B, T)$, where $B\colon P \to \mathbb{N}$ gives the seed count in each pit, $r_A, r_B \in \mathbb{N}$ are the two players' reserve counts, and $T \in \{A, B\}$ is the player to move.

Two invariants hold throughout play and require no assumption about the capture or relay rules. First, seeds are conserved:

$$\sum_{p \in P} B(p) + r_A + r_B = 64. \tag{1}$$

Second, a reserve never grows. Seeds leave a reserve only when introduced onto the board, and captured seeds are re-sown onto the board rather than returned to hand, so each reserve is bounded by its opening size $R \in \{18, 20\}$:

$$0 \le r_A \le R, \qquad 0 \le r_B \le R. \tag{2}$$

Definition 2 (Configuration space)**.** The configuration space C is the set of all positions satisfying (1) and (2).

We further write $S \subseteq C$ for the set of positions that are legal under the rules of the game, and $S_{\text{reach}} \subseteq S$ for those reachable from the initial setup under legal play. Thus

$$S_{\text{reach}} \subseteq S \subseteq C. \tag{3}$$

The state-space complexity proper is $|S_{\text{reach}}|$. Every bound proved in Section 3 is on $|C|$ and therefore, by (3), an upper bound on $|S_{\text{reach}}|$. We make no claim that C is tight: the gap is exactly the reachability question of Section 5.

2.3 Position adequacy

Counting positions, rather than play histories, presumes that a position carries all information that governs play. We state this as an explicit assumption rather than a theorem, because it can be verified only against a formal rule set.

Assumption 1 (Position adequacy)**.** The legality of a move and the resulting position depend only on the current position (B, r_A, r_B, T), and not on the history of play.

Three observations make Assumption 1 plausible. The phase is recovered from the reserves: a position is in the mtaji phase iff $r_A = r_B = 0$. The status of a kuu, namely whether it retains at least its opening seed count and so remains playable, is a function of the count $B(p)$ at the kuu pit. And the capture, relay, and direction rules, as commonly described, refer to current board occupancy and to the pit at which the last seed falls. Whether they refer to nothing else, for instance the type or direction of a preceding capture within the same move, is a property of the formal rules that must be checked.

Remark 1. The configuration count of Section 3 does not depend on Assumption 1. The assumption is required only to read that count as a bound on the game's state space rather than on board configurations. If the rules turned out to need history, the true state space could exceed $|C|$, and the position model would need a corresponding auxiliary component.

2.4 Moves and termination

A move selects a pit and a sowing direction and applies the relay-sowing and capture rules, yielding a successor position. Write $M(s)$ for the set of legal moves at s and $\delta(s, m)$ for the successor of s under $m \in M(s)$. A well-definedness question for Bao is whether a single move can fail to terminate through unbounded relay sowing. For the Zanzibar form this is studied probabilistically and remains open [6].

Remark 2 (Conditional totality of δ). The Malawian rules permit a player who judges that a non-capturing (*takata*) move will not terminate naturally to declare it a possible never-ending move and to halt after the current lap. If this rule is formalised as a bound on the number of sowing laps per move, then every move resolves to a unique successor and δ is total on $\{(s, m) : m \in M(s)\}$. This is a statement about the formal rule set, not a theorem of combinatorics, and it is not needed for the bound of Section 3. It is needed for the state graph $G = (S_{\text{reach}}, E)$, with $E = \{(s, \delta(s, m)) : m \in M(s)\}$, to be well-defined, which matters for the graph-theoretic analysis left to future work.

Theorem 1 (Finiteness)**.** *The configuration space C is finite.*

Proof. Each pit count lies in $\{0, \ldots, 64\}$, each reserve in $\{0, \ldots, R\}$ by (2), and T takes two values, with (1) constraining these further. A finite product of finite ranges is finite. □

3 The exact size of the configuration space

We compute $|C|$ exactly. The number of ways to place n indistinguishable seeds into k distinguishable pits is the stars-and-bars count $\binom{n+k-1}{k-1}$.

Lemma 1 (Mtaji phase)**.** *The number of mtaji-phase positions of a fixed side to move (those with $r_A = r_B = 0$) is*

$$N_{\text{mtaji}} = \binom{64 + 32 - 1}{32 - 1} = \binom{95}{31} = 9.901 \times 10^{24} \text{ (4 s.f.).}$$

Proof. With both reserves zero, (1) places all 64 seeds on the 32 pits, and the count is immediate. □

Lemma 2 (Namua phase)**.** *Let R be the opening reserve cap. The number of namua-phase positions of a fixed side to move (those with $(r_A, r_B) \neq (0, 0)$) is*

$$N_{\text{namua}}(R) = \sum_{\substack{0 \le a, b \le R \\ (a,b) \neq (0,0)}} \binom{(64 - a - b) + 32 - 1}{32 - 1}.$$

For $R = 20$, $N_{\text{namua}} = 7.740 \times 10^{25}$ (4 s.f.); for $R = 18$ the value is 7.738×10^{25}, differing from the $R = 20$ value already in the fourth significant figure (a relative difference of 1.9×10^{-4}).

Proof. Fix the side to move and a reserve pair (a, b) with $0 \le a, b \le R$ and $(a, b) \neq (0, 0)$. The board total is not a free variable: by (1) it is determined as $n = 64 - a - b$, and these n seeds are distributed over the 32 pits in $\binom{n+31}{31}$ ways. The map $(a, b, B) \mapsto (B, a, b, T)$ merely reassembles its arguments into a tuple, so it is injective onto the namua positions of the given side; distinct reserve pairs or distinct board vectors give distinct positions. Summing over all admissible (a, b) therefore counts each namua position exactly once. □

Proposition 1 (Phase partition)**.** *For a fixed side to move, the mtaji and namua position sets are disjoint and their union is the whole configuration space of that side.*

Proof. The phase of a position is a function of its reserves: mtaji iff $r_A = r_B = 0$, namua otherwise. The two sets are therefore disjoint by definition, and every position falls in exactly one of them. □

Theorem 2 (Configuration count)**.** *The configuration space has exact size*

$$|C| = N := 2\big(N_{\mathrm{mtaji}} + N_{\mathrm{namua}}(R)\big).$$

For $R = 20$, $N = 1.746 \times 10^{26}$ (4 s.f.). Consequently $|S_{\mathrm{reach}}| \le |S| \le N$.

Proof. By Proposition 1 the two phases partition the positions of a fixed side without overlap, so that side contributes $N_{\mathrm{mtaji}} + N_{\mathrm{namua}}(R)$ by Lemmas 1 and 2. Multiplying by the two values of T gives N, which is the exact cardinality of C. The inequality is (3). □

Remark 3. N is an exact count, computed from exact binomial arithmetic, not an order-of-magnitude estimate. What is loose is the inclusion $S_{\mathrm{reach}} \subseteq C$, not the value of N itself.

3.1 Reduction by symmetry, conditionally

Let ρ be the map that rotates the board by $180°$, exchanging the two players' territories, and flips the side to move. Whether ρ is an automorphism of the game depends on the capture and relay rules being invariant under it, which, like Assumption 1, can be confirmed only against the formal rules.

Assumption 2 (Player-swap invariance)**.** The move and capture rules are invariant under ρ.

Proposition 2 (Conditional symmetry reduction)**.** *Under Assumption 2, ρ is an involution on C that preserves legality and reachability, and the number of configurations up to ρ is $\frac{1}{2}(N+F)$, where F is the number of ρ-fixed configurations. Because ρ inverts the side to move, no position satisfies $\rho(s) = s$ — a fixed point would require $T = \overline{T}$ — so $F = 0$ and the reduced count is* exactly $N/2 = 8.730 \times 10^{25}$ *for $R = 20$. Equivalently, the factor of two from the side to move in Theorem 2 and the factor of one-half from ρ cancel, so the symmetry-reduced configuration count is simply the one-sided count $N_{\mathrm{mtaji}} + N_{\mathrm{namua}}(R)$.*

Proof. A symmetric two-player rule set invariant under ρ makes ρ a bijection of C carrying legal moves to legal moves, hence an involutive automorphism. By Burnside's lemma applied to the order-two group $\{\mathrm{id}, \rho\}$, the number of orbits is $\frac{1}{2}(|C| + F)$. Since ρ flips T, it sends every position to one with the opposite side to move and so fixes none; hence $F = 0$ and the number of orbits is exactly $|C|/2$. The vanishing of F is a property of ρ as a permutation of C and does not itself require Assumption 2; the assumption is what guarantees that ρ also carries legal moves to legal moves, so that the same exact halving descends to S and S_{reach}. □

A left-to-right reflection may furnish a further automorphism, since the two *kichwa* (head pits) and the two sowing directions are treated symmetrically, but whether it is exact depends on the finalised capture rule and is left open.

4 Bawo among other games

Table 1 places the bound of Theorem 2 alongside commonly cited state-space complexities of other games. Two cautions apply. First, the Bawo figure is an exact count of the configuration space, an upper bound on reachable positions, whereas the others are estimates of reachable positions, so the comparison is conservative in Bawo's direction. In particular, the placement of Bawo should be read as a ceiling: were its reachable fraction as small as is typical for mancala, $|S_{\mathrm{reach}}|$ could fall well below 10^{26}, conceivably toward the checkers range, so the table fixes only an upper position for Bawo and leaves the lower end undetermined. Second, cross-game comparisons of state-space complexity are only roughly meaningful, since games differ in how pieces enter and leave play.

The placement is consistent with the literature's account of Bao. Its renown rests less on a vast position count, since at $\sim 10^{26}$ the state space is large but far below Go's, than on its game-tree and decision complexity, driven by relay-sowing branching, very long games, and intricate evaluation. State-space complexity is, on this view, not the binding constraint on solving the game — which is a further reason to read the headline number as a coarse upper placement rather than a precise locus.

Game	State-space complexity ($\log_{10}$)
Awari / Oware (solved)	≈ 12
Checkers (solved)	≈ 21
Bawo (this paper, upper bound)	≈ 26
Chess	≈ 44
Go (19×19)	≈ 170

Table 1: Approximate state-space complexities. Non-Bawo figures are standard estimates of reachable positions [1]; the Bawo figure is the exact configuration count of Theorem 2, an upper bound on reachable positions whose true state-space complexity may lie substantially lower.

5 Toward the reachable state space

The state-space complexity proper is $|S_{\text{reach}}|$, strictly below N. Three structural constraints prune the space: reserves drain monotonically, a non-functioning kuu restricts the moves available, and most configurations are simply not reachable. Quantifying the gap requires an executable rule set and is the empirical core of the wider project.

As $N \sim 10^{26}$ precludes exhaustive enumeration, we propose statistical estimation over an engine that implements the model of Section 2:

- reachable-set size by capture and recapture (Lincoln–Petersen or Chao) over positions visited in independent random self-play games, deduplicated under the symmetry of Proposition 2 where Assumption 2 holds;
- game-tree complexity by Knuth's random-path estimator;
- average branching factor by sampling $|M(s)|$ along random play, separately per phase.

These estimators, and the executable rule specification they depend on, are the subject of companion work.

6 Related work

The rules of Zanzibar Bao were formalised by de Voogt [3], who also characterised expert play. Donkers, Uiterwijk, and de Voogt analysed Bao's computational properties and solved its novice version [4]; Donkers developed opponent-model search in this setting [5]. The termination of moves was studied by Kronenburg, Donkers, and de Voogt [6], and endgame-based solving by Kronenburg [7]. Learning approaches include the coevolutionary method of Conradie and Engelbrecht [2]. On the Malawian variant specifically, the ethnographic and descriptive record includes Sanderson [9] and Mwale [8]. To our knowledge no prior work reports a state-complexity analysis of the Malawian variant, which is the gap this paper begins to address.

7 Conclusion

We have given a finite-state model of Malawian Bawo that separates configuration, legal, and reachable states, and we have computed the exact size of the configuration space, approximately 1.7×10^{26}, which is an unconditional upper bound on the reachable state-space complexity. Under a stated and not-yet-verified player-swap invariance this reduces *exactly* to $N/2 \approx 8.7 \times 10^{25}$ up to symmetry, the halving being exact because the rotation has no fixed configuration. The model's remaining commitments, position adequacy and transition totality, are isolated as explicit assumptions that a formal rule specification must discharge, and the reachable state-space complexity, the quantity of ultimate interest, awaits that specification together with the sampling methodology of Section 5.

A Reproduction script

The figures of Section 3 are reproduced by the following self-contained Python script, which uses only exact integer arithmetic.

```
from math import comb

# Mtaji: all 64 seeds on 32 pits, fixed side to move.
Nmtaji = comb(64 + 32 - 1, 32 - 1)          # = C(95, 31)
```

```
def namua(R):
    # Namua: reserve pair (a, b) != (0, 0), board total fixed at 64 - a - b.
    return sum(comb(64 - a - b + 31, 31)
               for a in range(R + 1) for b in range(R + 1)
               if (a, b) != (0, 0))

for R in (20, 18):
    N = 2 * (Nmtaji + namua(R))             # |C|, both sides to move
    print(f"R={R}: Nmtaji={Nmtaji}  Nnamua={namua(R)}  |C|={N}  |C|/2={N // 2}")
```

References

[1] L. V. Allis. *Searching for Solutions in Games and Artificial Intelligence*. PhD thesis, University of Limburg, Maastricht, 1994.

[2] J. Conradie and A. P. Engelbrecht. Training Bao game-playing agents using coevolutionary particle swarm optimization. In *IEEE Symposium on Computational Intelligence and Games*, pp. 67–74, 2006.

[3] A. J. de Voogt. *Limits of the Mind: Towards a Characterisation of Bao Mastership*. CNWS, Leiden University, 1995.

[4] H. H. L. M. Donkers, J. W. H. M. Uiterwijk, and A. J. de Voogt. Mancala games, topics in mathematics and artificial intelligence. In *Step by Step: Proceedings of the 4th Colloquium Board Games in Academia*, 2002.

[5] H. H. L. M. Donkers. *Nosce Hostem: Searching with Opponent Models*. PhD thesis, Universiteit Maastricht, 2003.

[6] T. Kronenburg, H. H. L. M. Donkers, and A. J. de Voogt. Never-ending moves in Bao. *ICGA Journal*, 29(2):74–78, 2006.

[7] T. Kronenburg. *Towards a Quasi-Endgame-Based Bao Solver*. MSc thesis, Universiteit Maastricht, 2006.

[8] S. K. Mwale. Bao. *The Society of Malawi Journal*, 49(1):56–70, 1996.

[9] M. G. Sanderson. Bau (Swahili). *Journal of the Royal Anthropological Institute*, 43:726–736, 1913.